Wavemeter using a Michelson interferometer

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(Dated: February 15, 2024)

Having a precise knowledge of a laser wavelength is of crucial importance for optics experiments such as speed measurement,size measurement and spectroscopy. In this experiment the wavelength of a green He-Ne laser was measured using a two color Michelson interferometer leveraging a known red He-Ne laser wavelength. Recording the interference pattern of both beams with an oscilloscope, the frequency of each was obtained using a Fast Fourier Transform method. The result for the wavelength of the green laser is 545 ± 3 nm which coincide with the accepted value 543.5 ± 0.1 nm according to [2].

PACS numbers:

INTRODUCTION

The Michelson interferometer, created by physicist Albert A. Michelson, is an instrument that produces interference fringes by splitting a light beam, consider coherent and monochromatic, into two parts and then recombining them after they have traveled different optical paths. It is model that any phase difference between the two beams is due solely to the difference in the distances they travel. If we consider d as the relative path length change of the arms, and λ as the wavelength of the beam, then the optical intensity at the output of the interferometer is proportional to

$$|E_1|^2 + |E_2|^2 + 2\, E_1 \cdot E_2\, \cos\left(\frac{2\pi}{\lambda} d\right) \tag{1}$$

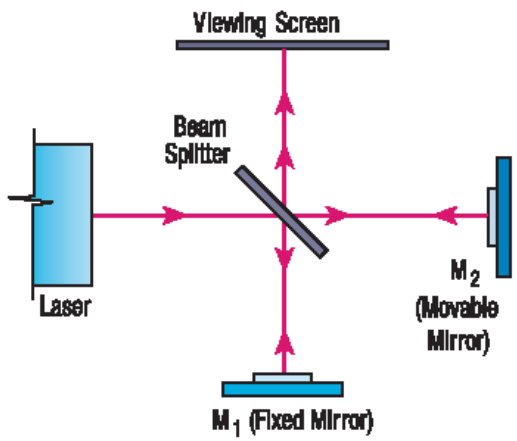

FIG. 1: Scheme of the Michelson interferometer. Reproduced from reference [1]

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If the movable mirror moves at a given speed, frequency of fringes can be predicted for a given lambda according to 1.The aim of this experiment is to calculate the wavelength of a green laser measuring the frequency ratio with a known red laser. The expression for the wavelength of the green laser λ_g in terms of the frequency ratio and the red reference wavelength λ_r is

$$\lambda_g = \frac{f_r}{f_g}\, \lambda_r \tag{2}$$

In this experiment we recorded the interference pattern of both lasers using an oscilloscope and acquired the frequency of each analyzing its frequency spectrum.

METHODS

A scheme of the set up can seen in Fig 2. The mounting consisted of two lasers of He-Ne,one red linearly polarized (RL) with a known wavelength and one green unpolarized (GL) with an unknown wavelength. The red laser was pointed directly into a polarizing beam splitter cube (BS1) whereas the green laser was pointed to a mirror (M1) and the reflected beam into the same beam splitter (BS1). The lasers were aligned collinear by adjusting the mirror (M1) checking overlap at near and far distances up to about 10 meters. After the beam splitter (BS1)

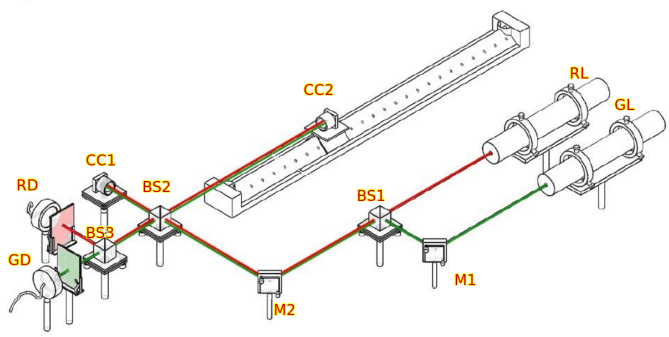

FIG. 2: Scheme of the double laser Michelson interferometer.

both lasers are reflected on a second mirror (M2) and introduced into de Michelson interferometer.

The moving corner cube (CC2) was supported over an air track and pushed at different speeds from 22 up to 35 cm/s .The green and red detectors, in the scheme (GD) and (RD) respectively, were connected to a TDS-3034C oscilloscope which enable us to record the interference pattern of both.

DATA & ANALYSIS

In Figure 3 we can appreciate the interference pattern generated by the red and green laser, each with its own frequency. This pattern was recorded at the output of the Michelson interferometer with an oscilloscope. For each waveform, the number of peaks was acquired by using a scipy function.

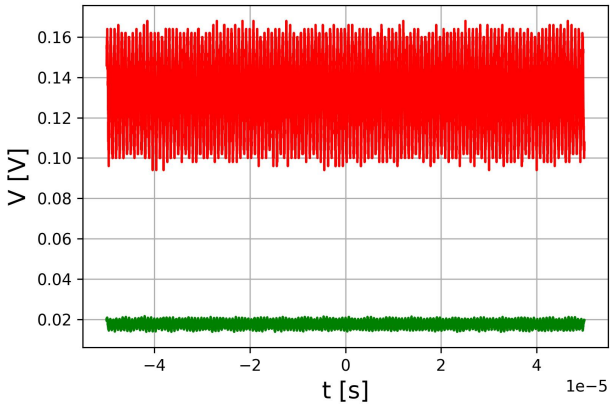

FIG. 3: Output of the Michelson interferometer recorded with an oscilloscope. Interference pattern for red and green laser can be appreciated. This graph correspond to a waveform of 202 ± 1 peaks for green and 178 ± 1 peaks for red.

For each measurement of the interference pattern the frequency of the signal was obtained by analyzing its Fourier Transform. This was accomplished numerically by using the Fast Fourier Transform method. In Fig 4 it is shown the frequency spectrum of the signal, the frequency of the highest peak is consider as the frequency of the laser beam. The uncertainty in frequency is taken as the half width of the narrow peak of each Fourier signal. In Table I it is tabulated the frequencies of each laser shown on Figure 3.

Laser	Frequency [kHz]
Green	1.510 ± 0.005
Red	1.300 ± 0.005

TABLE I: Frequency of each laser acquired by analyzing Fourier Transform of the interference pattern shown on Figure 3.

RESULTS

Taking the frequency ratio between the red and green laser and multiplying by the known wavelength of the red laser 632.8 ± 0.1 nm as described in equation (2). In Table II are tabulated the results for the wavelength of the green laser for each measurement.

green fringes	red fringes	frequency ratio	green wavelength [nm]
10	13	0.87 ± 0.05	548 ± 30
11	10	0.9 ± 0.1	570 ± 50
9	8	0.70 ± 0.06	440 ± 30
202	178	0.860 ± 0.006	545 ± 3

TABLE II: Results for green laser wavelength by measurement.

DISCUSSION

As it can be seen in Table II relative uncertainty depends strongly on the number of fringes per measurement. There is a measurement for which there are more red fringes than green ones, which makes no sense since is expected that the green laser has more frequency than that of the red. Because each measurement is subtended to different systematic uncertainty, it wouldn't be accurate to do statistics with them as they do not differ on an stochastic manner.

CONCLUSIONS

In this experiment the wavelength of a green He-Ne laser was measured using a two color Michelson interferometer leveraging a known red He-Ne laser wavelength. Recording the interference pattern of both beams with an oscilloscope, the frequency of each was obtained using a Fast Fourier Transform method. The systematic uncertainty on the measurement result to depend strongly on the number of fringes recorded, the reported value for the green laser wavelength is 545 ± 3 nm which coincide with the accepted value 543.5 ± 0.1 nm according to [2].

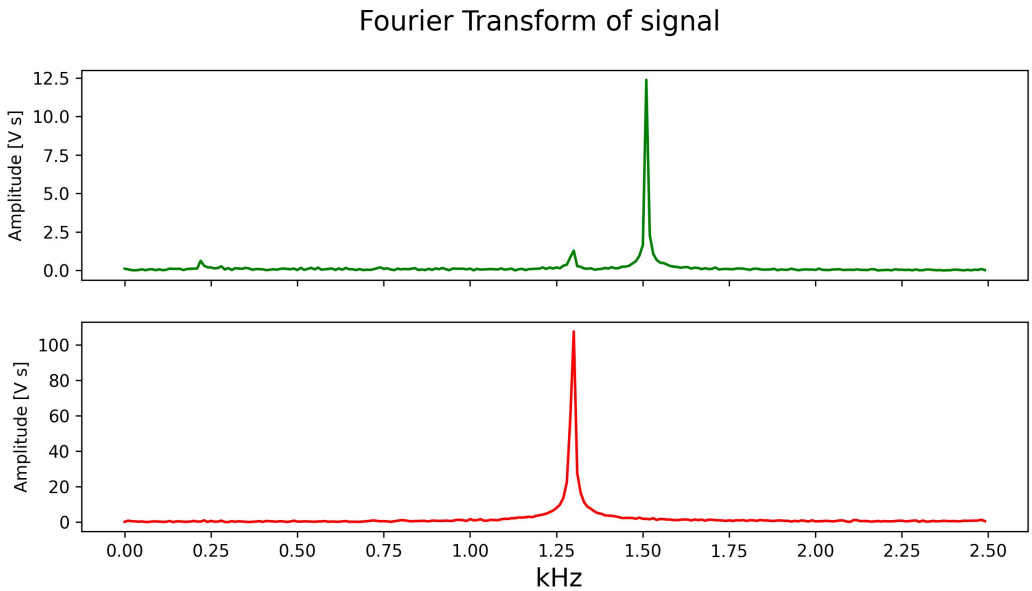

FIG. 4: Frequency spectrum for red and green laser. Obtained numerically using a Fast Fourier Transform method. The frequency of the highest peak is consider as the frequency of the laser beam. The uncertainty in frequency is taken as the half width of the narrow peak of each Fourier signal.

[1] Scheme of Michelson interferometer provided by PASCO.
[2] Value of reference for green He-Ne laser wavelenght.

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